Scaffold Fabrication for Cartilage Regeneration

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What is Articular Cartilage?

- Articular Cartilage is smooth white tissue that covers the ends of bones in joints.
  - Absorbs shock, reduces friction
  - Composed of dense extracellular matrix: water, collagen fibers, proteoglycans
  - Nonregenerative, avascular tissue

- Damage of cartilage leads to osteoarthritis
  - A type of arthritis, in which cartilage deteriorates
  - Bones rub against each other causing pain, stiffness, swelling, and reduced mobility
- ~530 million people worldwide suffer from osteoarthritis

Conventional and Current Treatment Options for Osteoarthritis

- **Lifestyle modifications**: weight management, exercise, joint protection (braces)
- **Medications**: Acetaminophen, NSAIDs (nonsteroidal anti-inflammatory drugs)
  - **Problem**: gastrointestinal toxicity + cardiovascular and renal risks
- **Surgical Options**:
  - Autologous chondrocyte implantation (ACI)
    1. Harvest chondrocytes from healthy cartilage
    2. Reimplant them into defects after expansion in in-vitro culture
    - **Problem**: 22% suffer hypertrophy (abnormal growth) after procedure
  - Bone marrow stimulation - microfracture, subchondral drilling
    - **Problem**: Regenerates fibrocartilage, not hyaline/articular cartilage
  - Autografts/allografts
    - **Problem**: donor site morbidity/limited cartilage supply/only for small defects
- **Emerging Treatments**:
  - Tissue engineering methods for scaffolds

Tissue Engineering Methods For Cartilage Regeneration

- Review 3 papers relevant to scaffolds for cartilage regeneration:
  1. Design and fabrication method of bi-layered fibrous scaffold for cartilage regeneration.
  2. Fabrication of chitosan/alginate/hydroxyapatite hybrid scaffolds using 3D printing and impregnating techniques for potential cartilage regeneration.
  3. Fabrication of Poly(Glycerol Sebacate)-Poly(ε-Caprolactone) Extrusion-Based scaffolds for cartilage regeneration.

Design and fabrication method of bi-layered fibrous scaffold for cartilage regeneration

The objective of the study was to **design and fabricate a bi-layered fibrous scaffold specifically for the regeneration of articular cartilage**. The scaffold aimed to **mimic the structural and functional properties of both cartilage** (chondral layer) **and bone** (subchondral layer) tissues.

- **Goals**: Enhancing Cartilage Regeneration., Facilitating Bone Tissue Formation, Evaluating Release Kinetics of Growth Factors
- **Scaffold features**: achieving high porosity, appropriate mechanical properties, and enhanced hydrophilicity

- **Materials:**
- Polycaprolactone (PCL), Cellulose Acetate (CA), Hydroxyapatite (HAP), Sodium Dodecyl Sulfate (SDS), Acetone, N.N-Dimethylformamide (DMF), Sodium Hydroxide (NaOH) Ethanol, and Growth Factor (TGFβ-3).

Design and fabrication method of bi-layered fibrous scaffold for cartilage regeneration

The scaffold was designed as a two-layer composite to simulate chondral (cartilage) and subchondral (bone) tissues.

Chondral Layer: Composed of PCL fibers with added cellulose, fabricated through a three-stage process:

1. Cryo-Electrospinning: PCL-CA scaffold was created using this method, which involves applying a high voltage to draw the polymer solution into fibers.
2. Deacetylation: The cellulose acetate was deacetylated to regenerate cellulose within the scaffold.
3. Surface Modification: The scaffold was modified to increase hydrophilicity.

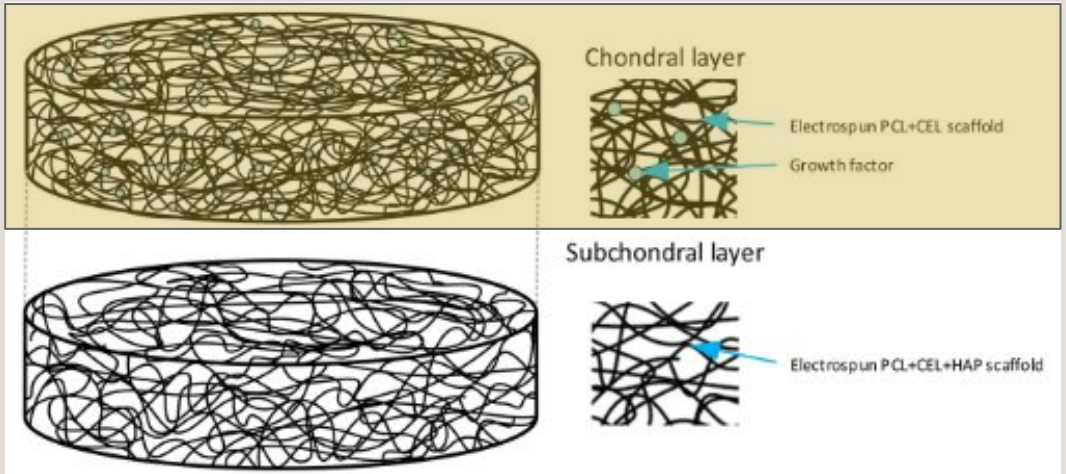


Fig. 1. Concept of construct for articular cartilage regeneration.

Design and fabrication method of bi-layered fibrous scaffold for cartilage regeneration

The scaffold was designed as a two-layer composite to simulate chondral (cartilage) and subchondral (bone) tissues.

Subchondral Layer: This layer included PCL, cellulose, and HAP, fabricated in two stages.

1. Cryo-Electrospinning: A PCL-CA-HAP scaffold was created.
2. Deacetylation: Similar to the chondral layer, cellulose was regenerated in the scaffold.

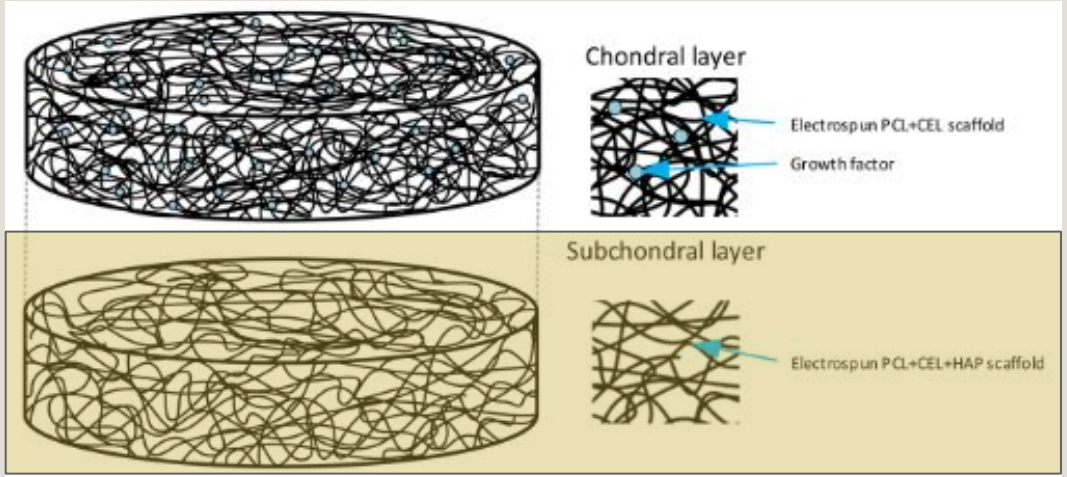


Fig. 1. Concept of construct for articular cartilage regeneration.

Design and fabrication method of bi-layered fibrous scaffold for cartilage regeneration

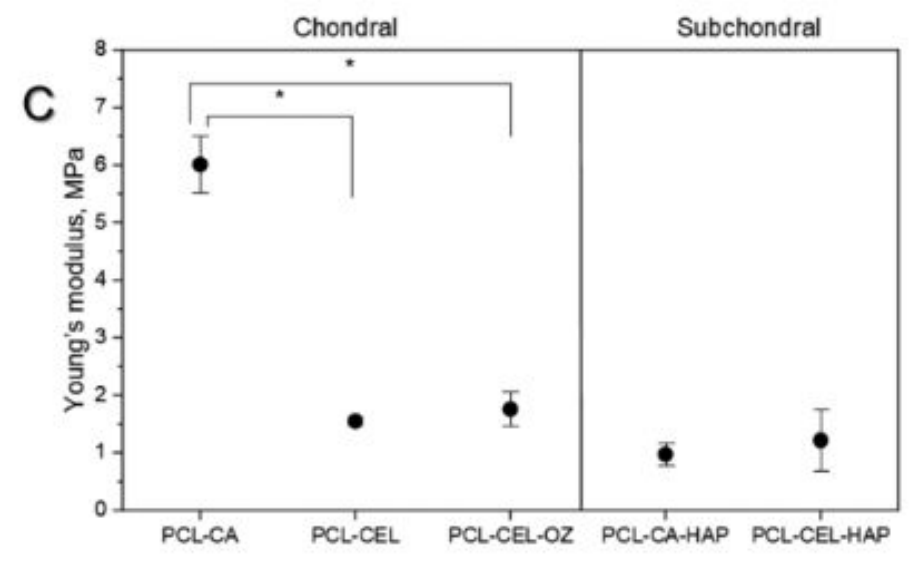
To evaluate the properties of the scaffold:

1. Surface Morphology Analysis: SEM → assessment of fiber diameter and distribution.
2. Wettability: water contact angle, absorption → hydrophilicity
3. Mechanical Testing: uniaxial Material Testing → tensile strength and young's modulus
4. Release of Growth Factors: ELISA → quantify the release of the chondrogenic growth factor TGFβ-3 from the scaffold over time → ability to deliver growth factors

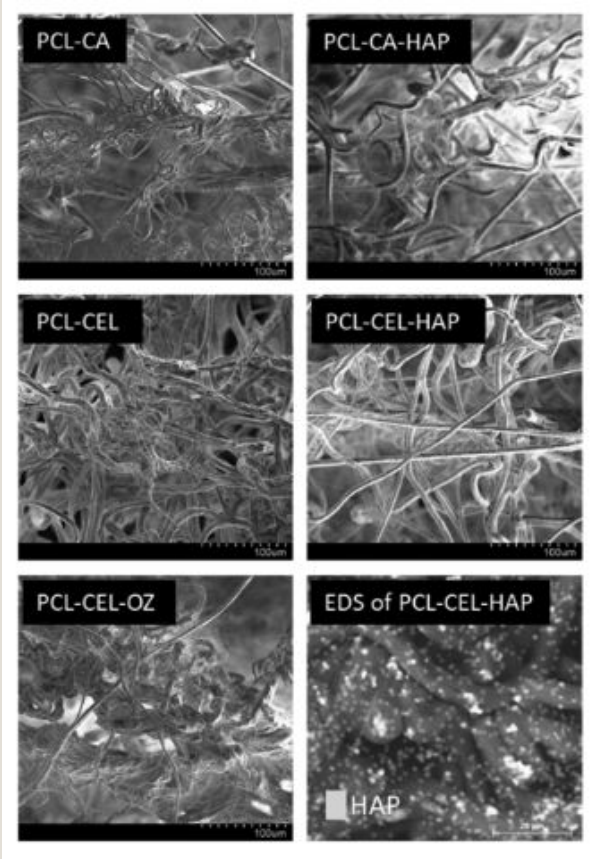
Design and fabrication method of bi-layered fibrous scaffold for cartilage regeneration

Wettability: Through the different steps of fabricating the scaffold, Deacetylation and ozonation both noticed an increase in wettability.

Mechanical Properties:



Morphology:



Design and fabrication method of bi-layered fibrous scaffold for cartilage regeneration

Release of Growth Factors over time:

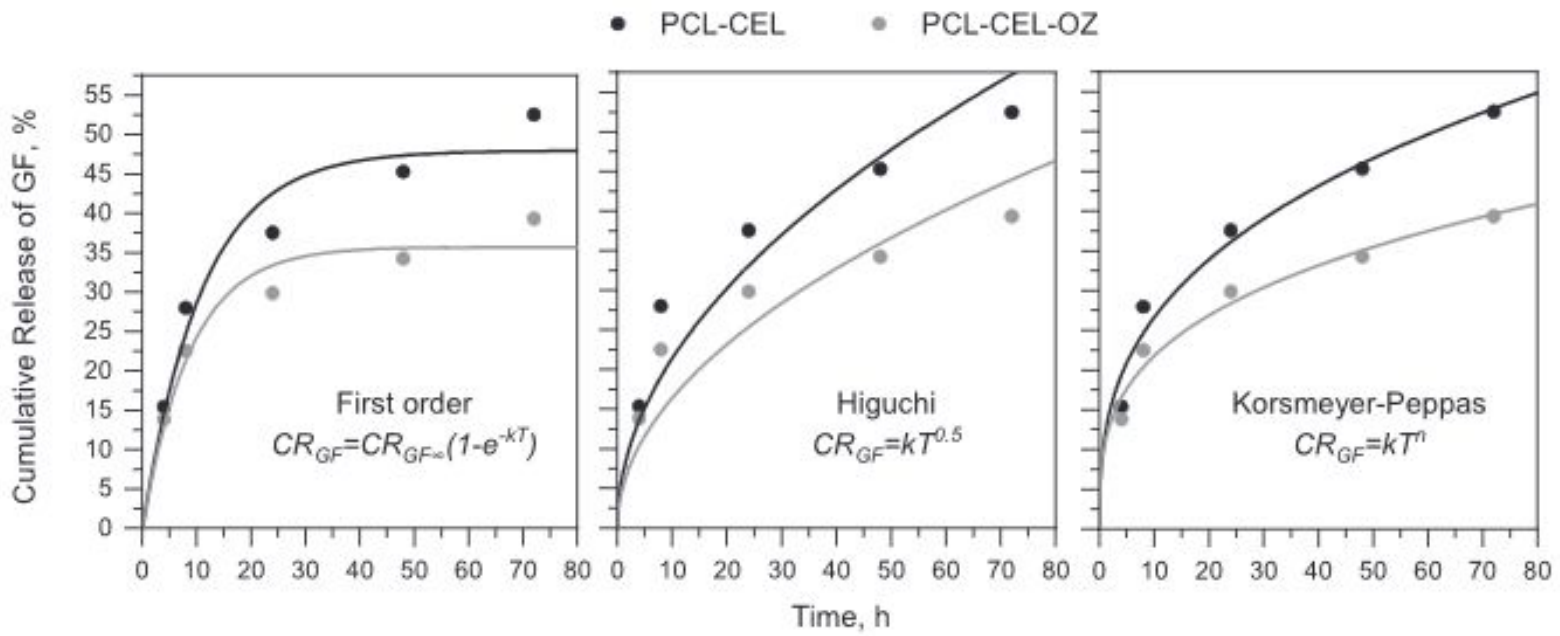


Fig. 6. Kinetic models of TGFβ-3 release from scaffolds.

Fabrication of chitosan/alginate/hydroxyapatite hybrid scaffolds using 3D printing and impregnating techniques for potential cartilage regeneration

- **AIM**: Improve biological properties of 3D printed chitosan scaffolds by combining with alginate and nHA

MATERIALS

- Chitosan: Natural polymer and hydrogel
  - Non-toxicity, biocompatibility
- Acetic acid: mixed with chitosan for printing
- Sodium alginate: when modified, can resemble natural ECM of cartilage
- Nano-hydroxyapatite: found in small quantities in cartilage
- Cross linking solutions:
  - Sodium hydroxide: crosslinks chitosan after printing
  - Calcium chloride: Crosslinks alginate to chitosan

METHODS

- 4 solutions for fabricating scaffolds:
  - C
  - CH
  - CHA
  - CAH
- Each solution was repeated with a different pore size
  - 2 mm and 3 mm porosity
- Printed with a 3D bioplotter at 37ºC and left to air dry for 24 hours

Fabrication of chitosan/alginate/hydroxyapatite hybrid scaffolds using 3D printing and impregnating techniques for potential cartilage regeneration

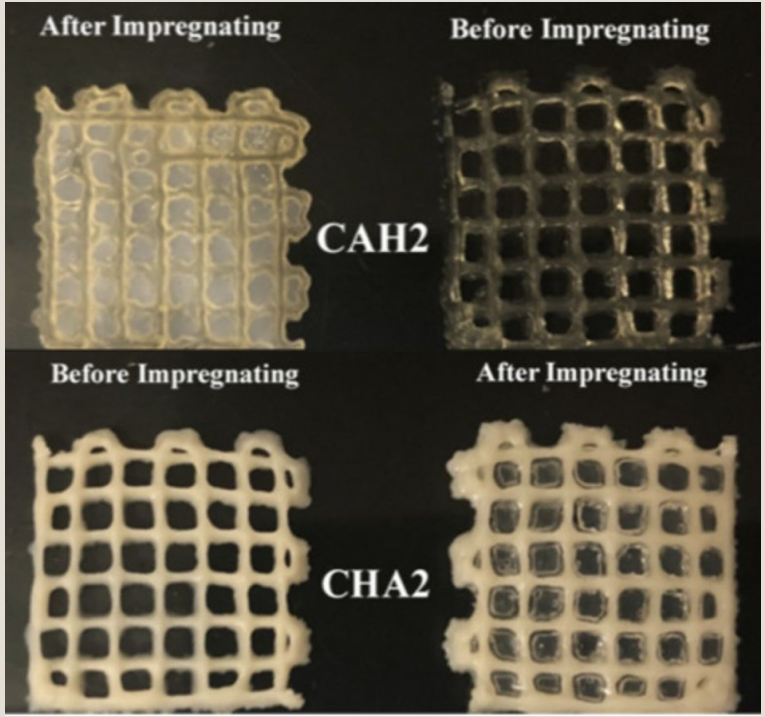
- Data table for 8 total experimental runs:

Sample Code	Concentration of Chitosan	Concentration of sodium alginate	Concentration of nHA	Porosity size
C	10%	0%	0%	2 or 3 mm
CH	10%	0%	10% in chitosan	2 or 3 mm
CHA	10%	3%	10% in chitosan	2 or 3 mm
CAH	10%	3%	10% in alginate	2 or 3 mm

Fabrication of chitosan/alginate/hydroxyapatite hybrid scaffolds using 3D printing and impregnating techniques for potential cartilage regeneration

METHODS (Cont.)

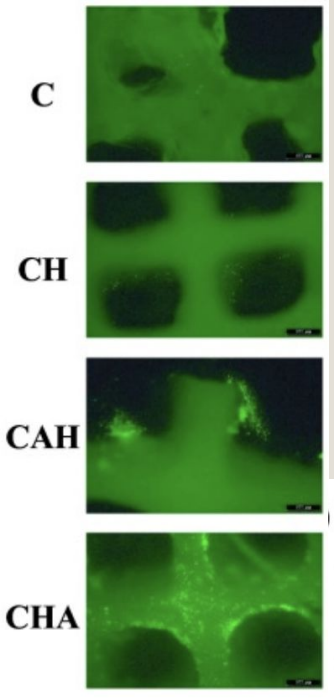
1. Live/Dead Assay: testing cell viability
   - ATDC5 chondrogenic cells were cultured in DMEM/F-12 & supplemented with FBS
   - After being counted, cells were seeded
   - Stained with calcein AM solution to visualize live cells; ethidium bromide to visualize dead cells
   - Used fluorescence microscopy
2. Mechanical testing
   - Placed under compressive load and tested elastic modulus
3. Degradation testing
   - Immersed in PBS solution, then incubated at 37°C
   - Dried and weighed every week for 4 weeks



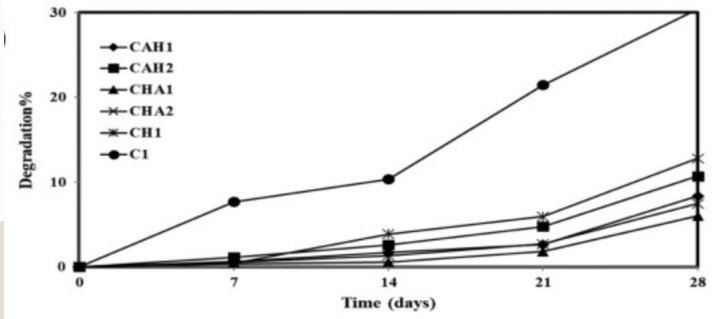
Fabrication of chitosan/alginate/hydroxyapatite hybrid scaffolds using 3D printing and impregnating techniques for potential cartilage regeneration

RESULTS AND DISCUSSION

- Live/Dead Assay
  - Not many live cells on C → adding nHA improved cell attachment
  - CAH - alginate/nHA solution filled pores, but strands were barely coated → less live cells on strands
  - CHA - high number of live cells on strands and pores
- Mechanical Testing
  - CAH & CHA - increasing pore size did not affect elastic modulus
  - Alginate/nHA solution did not change stiffness
    - But chitosan+nHA matrix increased stiffness (CH & CHA)



- Degradation Testing
  - Scaffolds containing nHA had lower degradation rate
  - nHA presence in chitosan scaffolds were more resistant than ones w/ alginate-nHA solution
  - Inclusion of alginate decreased degradation



Fabrication of Poly(Glycerol Sebacate)-Poly(ε-Caprolactone) Extrusion-Based Scaffolds for Cartilage Regeneration

MATERIALS

- **Poly(glycerol sebacate)** PGS:
  a. Biodegradable
  b. Biocompatible
  c. Flexibility
  d. Minimal Irritation
- **Poly(ε-Caprolactone)** PCL:
  a. Mechanical Stability
  b. Compatibility
- The combination improves **elasticity** and **strength**
  a. Resembling cartilage properties.

METHODS

- PGS and PCL were dissolved and mixed to create a uniform solution using chloroform(solvent).
- The solution was processed into **3D scaffolds** using a layer-by-layer manufacturing system.
- Pores were designed with precise size (650 μm inter-filament distance) to ensure oxygen and nutrient transportation.

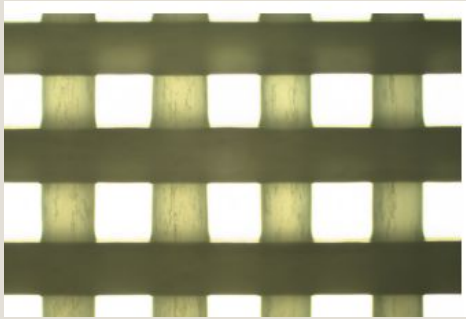


Figure 9: What the scaffold looks like after it is 3D printed.

Fabrication of Poly(Glycerol Sebacate)-Poly(ε-Caprolactone) Extrusion-Based Scaffolds for Cartilage Regeneration

RESULTS

Mechanical Properties:

- Scaffolds showed enhanced flexibility and mechanical strength.

Thermal Stability:

- Adding PGS increased thermal resistance → making the scaffold more stable.

Surface Topography:

- Filaments were aligned and interconnected → ensuring structural integrity.

DISCUSSION

- Scaffolds demonstrated high **biocompatibility** and **biodegradability** → breaking down naturally over 60 days without harmful byproducts.
- The elasticity supports medium-to-large deformations → minimizing tissue irritation during use.
- Controlled pore architecture promoted **vascularization** for improved tissue regeneration.

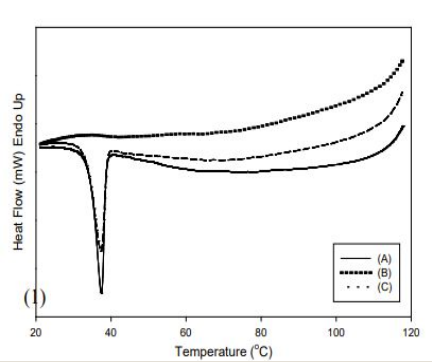


Figure 10: DSC thermo-graph for cooling style

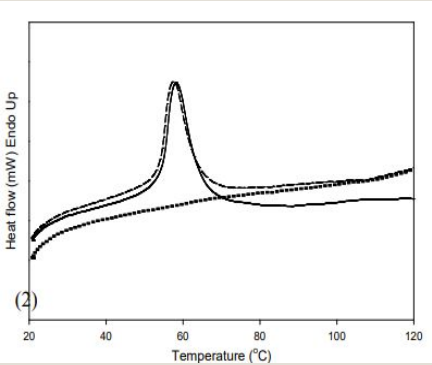


Figure 11: DSC thermo-graph for heating style

Recap

Materials

1 (Bi-layered Scaffold)

- Polycaprolactone (PCL)
- Cellulose
- Hydroxyapatite (HAP)

2 (Chitosan/Alginate Hybrid)

- Chitosan
- Alginate
- Nano-hydroxyapatite (nHA)

3 (PGS-PCL Scaffold)

- Polycaprolactone (PCL)
- Poly(glycerol sebacate) (PGS)

Techniques

- Uses **Cyro-electrospinning** (freezing + electrospinning)
  - Creates nanofiber w/ **high porosity**
  - Better **cell attachment** & **nutrient flow**
- Also goes through **Ozone treatment** to allow better water absorption and attract other functional groups as growth factors.

- Chitosan-based scaffolds were printed using **extrusion-based 3D printing**.
- Then infused w/ **alginate** and **nHA**.
  - Better **control over pore size**
  - Improved **interconnectivity**
  - Increased **cell visibility & attachment**

- The two polymers are dissolved and mixed to create a **homogeneous solution**.
- Then processed into scaffolds using a **layer by layer manufacturing system**.
- Pores were designed to be a specific size.
  - Promoted **vascularization**
- Scaffolds showed high:
  - **Biocompatibility**
  - **Biodegradability**

Conclusion and Future Outlook

- Three papers discuss different materials and methods for fabricating scaffolds for articular cartilage regeneration
  1. Cryo Electrospinning - PCL, CA, HA
  2. 3D printing
     a. PCL and PGS
     b. Chitosan, Alginate, nHA
- Takeaways:
  - Combining materials can yield scaffolds with greater mechanical properties
    - Could also open avenues to using different fabrication methods that would not be used typically
  - Post processing methods like ozone treatment can be used to improve biological properties
- Future Outlook:
  - The materials and methods utilized are promising for developing scaffolds closer to articular cartilage and for other applications

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